

Vaishnav Mahla

9408952402 | vdmahla1025@gmail.com | Surat, Gujarat, India

github.com/vaishnavmahla | linkedin.com/in/vaishnav-mahla | leetcode.com/vaishnav_mahla | codechef.com/vaishnav_1006

SUMMARY

Computer Science undergraduate with a strong foundation in C++, data structures & algorithms, and operating systems; comfortable working in Linux environments and debugging at the systems level. Focused on writing efficient, correct, and maintainable code, with hands-on experience building and optimizing real-time and backend systems.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

Surat, India

Bachelor of Technology in Computer Science Engineering; CGPA: 7.71/10

Expected 2028

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, SQL

Core CS: Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Object-Oriented Programming

Systems & Tools: Linux, Git, GitHub, VS Code, Postman

Databases: MySQL, MongoDB

Other: FastAPI, Node.js, Express.js, React, OpenCV

PROJECTS

Smart AI Traffic Management System | *C++, Python, OpenCV, YOLO, Arduino*

- Designed a modular real-time vehicle detection and signal-control pipeline, separating video capture, inference, and hardware-control logic into independent components to simplify testing and debugging.
- Implemented real-time object detection and vehicle counting using OpenCV and YOLO, processing live video frames under latency constraints to keep signal decisions responsive.
- Built a rule-based decision layer that dynamically adjusts signal timing based on live vehicle density per lane, replacing static fixed-interval switching.
- Profiled the inference loop to identify frame-processing bottlenecks and reduced redundant computation, improving throughput without sacrificing detection accuracy.
- Interfaced the software pipeline with Arduino over serial communication to control traffic signal hardware, handling edge cases such as dropped frames and sensor noise.

IPL Match Predictor | *FastAPI, React, Machine Learning*

- Architected a backend prediction pipeline in FastAPI that ingests historical match data, applies feature engineering (team form, venue, toss outcome), and serves win-probability predictions through a REST API.
- Structured the codebase into clearly separated data-preprocessing, model-inference, and API layers to keep the pipeline testable and easy to extend with new features.
- Deployed the trained model behind versioned REST endpoints and connected a React frontend to consume predictions, decoupling model updates from client-side changes.
- Wrote input-validation and error-handling logic to keep the API resilient to malformed or missing match data.

Library Management System | *Node.js, Express, MongoDB, React*

- Designed a normalized MongoDB schema and modular Express backend exposing REST endpoints for CRUD operations on books, members, and transactions.
- Implemented token-based authentication and role-based access control to separate admin and user privileges across API routes.
- Structured the backend into distinct controller, route, and model layers to isolate business logic from data access and simplify future maintenance.
- Handled concurrent borrow/return operations with validation logic to prevent inconsistent book-availability states.

ACHIEVEMENTS

Solved 320+ algorithmic and data-structures problems across LeetCode and CodeChef, reflecting consistent, self-driven practice in problem-solving.

LeetCode: Solved 77 problems (38 Easy, 39 Medium) covering arrays, strings, trees, graphs, and dynamic programming.

CodeChef: Solved 323+ problems across contests, with regular participation in competitive rated rounds.